Chapter 7. Hypothesis Testing

Logic of tests, likelihood ratio tests, large-sample approximations

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7.1 The logic of testing

If the observed result would be very unlikely under a proposed model, we reject the model in favour of an alternative: **statistical hypothesis testing**.

Example 7.1

A standard painkiller acts in $N(20, 4^2)$ minutes; a new drug is claimed faster. With $n = 64$, $\bar{X} \sim N(\mu, 0.5^2)$; test $H_1 : \mu < 20$ against $H_0 : \mu = 20$.
Small $\bar{X}$ is hard to explain under H_0:

$$P(\bar{X} \le 19 \mid H_0) = P\Big(\frac{\bar{X} - 20}{0.5} \le -2\Big) = 0.023 :$$

about twice in a hundred trials under H_0: strong evidence against it.

H_0 is the **null hypothesis**, H_1 the **alternative**. The set C of samples rejecting H_0 is the **rejection** (critical) region, its complement the **acceptance region**; a statistic describing C (here $\bar{X}$) is a **test statistic**. Errors:

	H_0 true	H_1 true
accept H_0	correct	type II error
reject H_0	type I error	correct

We bound the type I error probability by a small **significance level** α (0.10, 0.05, 0.01): $P(\text{reject} \mid H_0) \le \alpha$ defines a **level-α test**. To minimize type II error, make the rejection region as large as the level allows.

Example 7.2

"$\bar{X} \le 19$" (probability 0.023 under H_0) and "$\bar{X} \le 20 - z_{0.05} \cdot 0.5 = 19.1775$" (exactly 0.05) are both level 0.05. Under μ,

$$P_\mu(\bar{X} \le 19) = \Phi(38 - 2\mu), \qquad P_\mu(\bar{X} \le 19.1775) = \Phi(38.355 - 2\mu).$$

The second is larger for every $\mu < 20$ (smaller type II error); generally the region is $\bar{X} \le 20 - z_\alpha \cdot 0.5$.

Example 7.3 (Randomization)

Poisson(θ) sample, $n = 50$; test $H_0 : \theta = 0.1$ vs $H_1 : \theta < 0.1$ at $\alpha = 0.05$. Small $S = \sum X_i$ speaks against H_0, and $S \sim \text{Poisson}(5)$ under H_0:

$$P_{0.1}(S \le 0) = 0.0067, \qquad P_{0.1}(S \le 1) = 0.0404, \qquad P_{0.1}(S \le 2) = 0.1247.$$

No region $S \le k$ has probability exactly 0.05; reject for $S \le 1$, with probability γ for $S = 2$, accept for $S \ge 3$:

$$0.0404 + \gamma \cdot P_{0.1}(S = 2) = 0.05, \qquad \gamma = \frac{0.05 - 0.0404}{0.0842} = 0.114.$$

a **randomized test**.

General formulation. Let the model be $\{f(x;\theta) : \theta \in \Omega\}$ and test

$$H_0 : \theta \in \Omega_0 \quad \text{vs} \quad H_1 : \theta \in \Omega_1, \qquad \Omega_0 \cap \Omega_1 = \varnothing,\ \Omega_0 \cup \Omega_1 = \Omega.$$

A (possibly randomized) test is a **test function** $\phi(\mathbf{x}) \in [0,1]$, the probability of rejecting at $\mathbf{X} = \mathbf{x}$ ($\phi = \mathbf{1}_C$ if nonrandomized). Its **power function** is $\gamma_\phi(\theta) = E_\theta \phi(\mathbf{X})$, its **size** $\max_{\theta \in \Omega_0} E_\theta \phi(\mathbf{X})$; level α means size $\le \alpha$. We seek size-α tests with maximal power on Ω_1.

Example 7.4 (Composite null)

In Example 7.1 with $H_0 : \mu \ge 20$: $P_\mu(\bar{X} \le c) = \Phi\big((c-\mu)/0.5\big)$ decreases in μ, so the size is attained at $\mu = 20$, giving the same region $\bar{X} \le 20 - 0.5 z_\alpha$, with power $\gamma(\mu) = \Phi(38.355 - 2\mu)$ ($\alpha = 0.05$). In Example 7.3

$$\gamma(\theta) = \sum_{x=0}^{1} e^{-50\theta} \frac{(50\theta)^x}{x!} + 0.114\, e^{-50\theta} \frac{(50\theta)^2}{2},$$

also decreasing in θ.

7.2 Likelihood ratio tests

Compare the best fits under each hypothesis:

$$\frac{\max_{\theta \in \Omega_1} L(\theta; \mathbf{x})}{\max_{\theta \in \Omega_0} L(\theta; \mathbf{x})}.$$

Large values are evidence against H_0. Since $\max_\Omega L = \max\{\max_{\Omega_0} L, \max_{\Omega_1} L\}$,

$$\frac{\max_\Omega L}{\max_{\Omega_0} L} = \max\left\{1, \frac{\max_{\Omega_1} L}{\max_{\Omega_0} L}\right\},$$

which is monotone in the first ratio; so use the full maximum, with MLEs $\hat\theta^\Omega$, $\hat\theta^{\Omega_0}$:

Definition (Likelihood ratio test)

The **likelihood ratio test** (LRT) of level α rejects H_0 when

$$2\{l(\hat\theta^\Omega) - l(\hat\theta^{\Omega_0})\} \ge c, \qquad \text{with } c \text{ chosen so that} \quad \max_{\theta \in \Omega_0} P_\theta(\text{reject}) = \alpha$$

(randomizing if necessary).

Example 7.5 (Two-sided t test)

$N(\mu, \sigma^2)$, $n \ge 2$; $H_0 : \mu = \mu_0$ vs $H_1 : \mu \ne \mu_0$. With

$$l(\mu, \sigma^2) = -\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 - \frac{n}{2} \log \sigma^2 - \frac{n}{2} \log 2\pi,$$

$\hat\theta^\Omega = (\bar{x}, \hat\sigma^2)$, $\hat\sigma^2 = \frac{1}{n}\sum (x_i - \bar{x})^2$, and $\hat\theta^{\Omega_0} = (\mu_0, \hat\sigma_0^2)$, $\hat\sigma_0^2 = \hat\sigma^2 + (\bar{x} - \mu_0)^2$. Hence

$$2\{l(\hat\theta^\Omega) - l(\hat\theta^{\Omega_0})\} = n \log \frac{\hat\sigma_0^2}{\hat\sigma^2} = n \log\left(1 + \frac{(\bar{x} - \mu_0)^2}{\hat\sigma^2}\right),$$

increasing in $|\bar{x} - \mu_0|/(s/\sqrt{n})$. Since $\sqrt{n}(\bar{X} - \mu_0)/S \sim t(n-1)$ under H_0, the LRT is the **two-sided t test**:

$$\text{reject } H_0 \iff \left|\frac{\bar{X} - \mu_0}{S/\sqrt{n}}\right| \ge t_{\alpha/2}(n-1).$$

Example 7.6 (One-sided t test)

$H_0 : \mu \le \mu_0$ vs $H_1 : \mu > \mu_0$. Over Ω_0, $\hat\mu^{\Omega_0} = \min(\bar x, \mu_0)$ and $\hat\sigma^{2,\Omega_0} = \frac{1}{n}\sum(x_i - \hat\mu^{\Omega_0})^2$, so

$$2\{l(\hat\theta^{\Omega}) - l(\hat\theta^{\Omega_0})\} = n\log\Big(1 + \frac{(\bar x - \min(\bar x, \mu_0))^2}{\hat\sigma^2}\Big),$$

0 for $\bar x \le \mu_0$, increasing in $(\bar x - \mu_0)/(s/\sqrt{n})$ otherwise: the region is $\{\sqrt{n}(\bar x - \mu_0)/s \ge c\}$, $c > 0$. With independent $Z = \sqrt{n}(\bar X - \mu)/\sigma \sim N(0,1)$, $V = (n-1)S^2/\sigma^2 \sim \chi^2(n-1)$,

$$P_{\mu,\sigma^2}\Big(\frac{\bar X - \mu_0}{S/\sqrt{n}} \ge c\Big) = P\Big(c\sqrt{V/(n-1)} - Z \le \frac{\mu - \mu_0}{\sigma/\sqrt{n}}\Big),$$

increasing in μ, so the size is at $\mu = \mu_0$: the **one-sided t test** rejects when $\sqrt{n}(\bar X - \mu_0)/S \ge t_\alpha(n-1)$.

Example 7.7 (Exponential mean)

$\mathrm{Exp}(\theta)$, $H_0 : \theta = \theta_0$ vs $H_1 : \theta \ne \theta_0$: $\hat\theta^{\Omega} = \bar x$ and

$$2\{l(\hat\theta^{\Omega}) - l(\theta_0)\} = 2n\Big(\frac{\bar x}{\theta_0} - 1 - \log\frac{\bar x}{\theta_0}\Big).$$

As $t - 1 - \log t$ decreases on $(0,1)$ and increases on $(1,\infty)$, the region is

$$\frac{\bar x}{\theta_0} \le c_1 \quad \text{or} \quad \frac{\bar x}{\theta_0} \ge c_2, \qquad c_1 - \log c_1 = c_2 - \log c_2.$$

Under H_0, $2n\bar X/\theta_0 \sim \chi^2(2n)$, so $\int_{2nc_1}^{2nc_2} f_{\chi^2(2n)} = 1 - \alpha$ determines c_1, c_2. For large n, $c_{1,2} \approx 1 \mp z_{\alpha/2}/\sqrt{n}$: approximately "$n(\bar X/\theta_0 - 1)^2 \ge \chi^2_\alpha(1)$".

Example 7.8 (Two-sample t test)

Independent $N(\mu_i, \sigma^2)$ samples, $i = 1, 2$; $H_0 : \mu_1 = \mu_2$. Under Ω, $\hat\mu_i = \bar x_i$, $\hat\sigma^2 = \sum_{i,j}(x_{ij} - \bar x_i)^2/(n_1 + n_2)$; under Ω_0, $\hat\mu = (n_1\bar x_1 + n_2\bar x_2)/(n_1 + n_2)$ and $\hat\sigma_0^2 = \hat\sigma^2 + \frac{n_1 n_2}{(n_1+n_2)^2}(\bar x_1 - \bar x_2)^2$:

$$2\{l(\hat\theta^{\Omega}) - l(\hat\theta^{\Omega_0})\} = (n_1 + n_2)\log\Big(1 + \frac{n_1 n_2}{(n_1 + n_2)^2}\frac{(\bar x_1 - \bar x_2)^2}{\hat\sigma^2}\Big),$$

increasing in $|\bar x_1 - \bar x_2|/(s_p\sqrt{1/n_1 + 1/n_2})$, $s_p^2 = \sum_{i,j}(x_{ij} - \bar x_i)^2/(n_1 + n_2 - 2)$, which is $t(n_1 + n_2 - 2)$ under H_0 (Corollary 4.10 in Volume I). Reject when

$$\frac{|\bar X_1 - \bar X_2|}{S_p\sqrt{1/n_1 + 1/n_2}} \ge t_{\alpha/2}(n_1 + n_2 - 2).$$

Example 7.9 (Comparing two variances)

$N(\mu_i, \sigma_i^2)$ samples; $H_0 : \sigma_1^2 = \sigma_2^2$. With $\hat\sigma_i^2 = \frac{1}{n_i}\sum_j(x_{ij} - \bar x_i)^2$ and pooled $\hat\sigma^2 = (n_1\hat\sigma_1^2 + n_2\hat\sigma_2^2)/(n_1 + n_2)$,

$$2\{l(\hat\theta^{\Omega}) - l(\hat\theta^{\Omega_0})\} = (n_1 + n_2)\log\hat\sigma^2 - n_1\log\hat\sigma_1^2 - n_2\log\hat\sigma_2^2.$$

With $t = n_1\hat\sigma_1^2/(n_1\hat\sigma_1^2 + n_2\hat\sigma_2^2)$ this is $-n_1\log t - n_2\log(1-t) + \text{const}$, minimal at $t = n_1/(n_1 + n_2)$; t increases in $F = S_1^2/S_2^2 \sim F(n_1 - 1, n_2 - 1)$ under H_0, so the region is $F \le d_1$ or $F \ge d_2$

($\int_{d_1}^{d_2} f_F = 1-\alpha$, equal statistic at d_1, d_2). In practice $d_1 = F_{1-\alpha/2}$, $d_2 = F_{\alpha/2}$.

Example 7.10 (Poisson, one-sided)

As in Example 7.3, $H_0 : \theta \geq 0.1$ vs $H_1 : \theta < 0.1$: $\hat\theta^{\Omega} = \bar x$, $\hat\theta^{\Omega_0} = \max(\bar x, 0.1)$, and

$$2\{l(\hat\theta^{\Omega}) - l(\hat\theta^{\Omega_0})\} = 2n\{\bar x \log \bar x - \bar x(1+\log 0.1) + 0.1\}\mathbf{1}\{\bar x < 0.1\},$$

decreasing in $\bar x \leq 0.1$: reject for small S. As $P_\theta(S \leq k) = \int_{n\theta}^{\infty} z^k e^{-z}/k!\, dz$ decreases in θ, the size is at $\theta = 0.1$, and Example 7.3 is the level-0.05 LRT.

7.3 Large-sample approximations

Without exact null laws, use MLE asymptotics: $\Omega \subseteq \mathbb{R}^k$ open, $\bar l = l/n$, Fisher information $I(\theta)$, and $\sqrt{n}(\hat\theta_n - \theta) \approx I(\theta)^{-1}\sqrt{n}\,\dot{\bar l}(\theta)$ (Chapter 6).

Simple null hypothesis. $H_0 : \theta = \theta_0$. Expanding about $\hat\theta_n$ ($\dot l(\hat\theta_n) = 0$),

$$l(\theta_0) = l(\hat\theta_n) + \tfrac{1}{2}(\theta_0 - \hat\theta_n)'\ddot l(\theta_n^*)(\theta_0 - \hat\theta_n),$$

with $-\frac{1}{n}\ddot l(\theta_n^*) \xrightarrow{p} I(\theta_0)$ under H_0. Hence

$$2\{l(\hat\theta_n) - l(\theta_0)\} \approx \sqrt{n}(\hat\theta_n - \theta_0)' I(\theta_0)\sqrt{n}(\hat\theta_n - \theta_0).$$

and $\sqrt{n}(\hat\theta_n - \theta_0) \xrightarrow{d} \mathbf{Z} \sim N_k(\mathbf{0}, I(\theta_0)^{-1})$ gives the limit $\mathbf{Z}' I(\theta_0)\mathbf{Z} \sim \chi^2(k)$ (Theorem 4.18 in Volume I).

Theorem 7.1 (Simple null hypothesis)

Under (R0)–(R7) of Theorem 6.13, for $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$, under H_0:

(a) $2\{l(\hat\theta_n) - l(\theta_0)\} \xrightarrow{d} \chi^2(k)$; rejecting when it is $\geq \chi^2_\alpha(k)$ has asymptotic level α.

(b) $2\{l(\hat\theta_n) - l(\theta_0)\} = W_n(\theta_0) + o_p(1)$, where $W_n(\theta_0) = n(\hat\theta_n - \theta_0)' I(\theta_0)(\hat\theta_n - \theta_0)$.

(c) $2\{l(\hat\theta_n) - l(\theta_0)\} = R_n(\theta_0) + o_p(1)$, where $R_n(\theta_0) = n\,\dot{\bar l}(\theta_0)' I(\theta_0)^{-1}\dot{\bar l}(\theta_0)$.

Sketch. (a), (b) as above (remainder as in Theorem 6.13). (c): substitute $\sqrt{n}(\hat\theta_n - \theta_0) = I(\theta_0)^{-1}\sqrt{n}\,\dot{\bar l}(\theta_0) + o_p(1)$ into W_n. ■

W_n is **Wald's statistic**, R_n **Rao's (score) statistic**; all three are asymptotically $\chi^2(k)$. The score test needs only θ_0.

Example 7.11 (Exponential)

Exp(θ), $H_0 : \theta = \theta_0$: $\dot{\bar l}(\theta) = -\frac{1}{\theta} + \frac{\bar x}{\theta^2}$, $\hat\theta = \bar x$, $I(\theta) = 1/\theta^2$:

$$2\{l(\hat\theta) - l(\theta_0)\} = 2n\Big(\frac{\bar x}{\theta_0} - 1 - \log\frac{\bar x}{\theta_0}\Big), \qquad W_n = \frac{n(\bar x - \theta_0)^2}{\bar x^2}, \qquad R_n = \frac{n(\bar x - \theta_0)^2}{\theta_0^2}.$$

each compared with $\chi^2_\alpha(1)$.

Example 7.12 (Multinomial goodness of fit)

$\mathbf{Z}_i$ i.i.d. Multi$(1, \mathbf{p})$, counts $X_j = \sum_i Z_{ij}$, $H_0 : \mathbf{p} = \mathbf{p}_0$. With $\theta = (p_1, \ldots, p_r)'$, $r = k-1$, $\theta_\cdot = \sum_{j \le r} \theta_j = 1 - p_k$,

$$l(\theta) = \sum_{j \le r} x_j \log \theta_j + x_k \log(1 - \theta_\cdot), \qquad \hat{\theta}_j = x_j / n,$$

$$I(\theta) = \operatorname{diag}(1/\theta_j) + \frac{1}{p_k} \mathbf{1}\mathbf{1}'.$$

One computes $I(\theta_0)(\hat{\theta} - \theta_0) = \dot{l}(\theta_0)$, with j-th entry $\frac{x_j - n p_{0j}}{n p_{0j}} - \frac{x_k - n p_{0k}}{n p_{0k}}$, so

$$W_n(\theta_0) = R_n(\theta_0) = \sum_{j=1}^{k} \frac{(X_j - n p_{0j})^2}{n p_{0j}},$$

Pearson's chi-square; the LRT statistic is $2 \sum_j X_j \log \frac{X_j}{n p_{0j}}$. All are asymptotically $\chi^2(k-1)$ (cf. Example 5.12 in Volume I).

Composite null hypotheses. Write $\theta = (\xi', \eta')'$, $\xi \in \mathbb{R}^{k_1}$, $\eta \in \mathbb{R}^{k_0}$, $k_0 + k_1 = k$, $H_0 : \xi = \xi_0$, and partition

$$\dot{l} = \begin{pmatrix} \dot{l}_1 \\ \dot{l}_2 \end{pmatrix}, \qquad I = \begin{pmatrix} I_{11} & I_{12} \\ I_{21} & I_{22} \end{pmatrix}, \qquad I_{11 \cdot 2} = I_{11} - I_{12} I_{22}^{-1} I_{21}.$$

with $\hat{\theta}_n$ the unrestricted MLE and $\hat{\theta}_n^0 = (\xi_0', \hat{\eta}_n^{0\prime})'$ the MLE under H_0.

Theorem 7.2 (Composite null hypothesis)

Under Theorem 6.13's conditions, if $H_0 : \xi = \xi_0$ holds with $\theta_0 = (\xi_0', \eta_0')'$,

(a) $2\{l(\hat{\theta}_n) - l(\hat{\theta}_n^0)\} \xrightarrow{d} \chi^2(k - k_0)$;

(b) $2\{l(\hat{\theta}_n) - l(\hat{\theta}_n^0)\} = n(\hat{\theta}_n - \hat{\theta}_n^0)' I(\theta_0)(\hat{\theta}_n - \hat{\theta}_n^0) + o_p(1)$;

(c) $2\{l(\hat{\theta}_n) - l(\hat{\theta}_n^0)\} = n\, \dot{l}(\hat{\theta}_n^0)' I(\theta_0)^{-1} \dot{l}(\hat{\theta}_n^0) + o_p(1)$.

Sketch. As in Theorem 7.1,

$$2\{l(\hat{\theta}_n) - l(\theta_0)\} \approx \sqrt{n}\, \dot{l}(\theta_0)' I^{-1} \sqrt{n}\, \dot{l}(\theta_0).$$

and for the submodel with $\xi = \xi_0$ fixed,

$$2\{l(\hat{\theta}_n^0) - l(\theta_0)\} \approx \sqrt{n}\, \dot{l}_2(\theta_0)' I_{22}^{-1} \sqrt{n}\, \dot{l}_2(\theta_0).$$

Subtracting and using

$$\begin{pmatrix} I_{11} & I_{12} \\ I_{21} & I_{22} \end{pmatrix}^{-1} = \begin{pmatrix} I_{11\cdot 2}^{-1} & -I_{11\cdot 2}^{-1} I_{12} I_{22}^{-1} \\ -I_{22}^{-1} I_{21} I_{11\cdot 2}^{-1} & I_{22}^{-1} + I_{22}^{-1} I_{21} I_{11\cdot 2}^{-1} I_{12} I_{22}^{-1} \end{pmatrix},$$

gives

$$2\{l(\hat{\theta}_n) - l(\hat{\theta}_n^0)\} \approx \sqrt{n}\, \mathbf{Z}_n' I_{11\cdot 2}^{-1} \sqrt{n}\, \mathbf{Z}_n, \qquad \mathbf{Z}_n = \dot{l}_1(\theta_0) - I_{12} I_{22}^{-1} \dot{l}_2(\theta_0).$$

By the CLT $\sqrt{n}\, \mathbf{Z}_n \xrightarrow{d} N_{k_1}(\mathbf{0}, I_{11\cdot 2})$, so the limit is $\chi^2(k_1)$: (a). For (b), $\hat{\eta}_n^0 - \eta_0 \approx I_{22}^{-1} \dot{l}_2(\theta_0)$ and $I_{11\cdot 2}(\hat{\xi}_n - \xi_0) \approx \mathbf{Z}_n$ give $n(\hat{\theta}_n - \hat{\theta}_n^0)' I (\hat{\theta}_n - \hat{\theta}_n^0) \approx n \mathbf{Z}_n' I_{11\cdot 2}^{-1} \mathbf{Z}_n$;

(c) is similar. ∎

With $I(\hat\theta_n^0)$ for $I(\theta_0)$, (b) and (c) are the composite Wald and Rao statistics. Smooth constraints $H_0: g_1(\theta)=\cdots=g_{k_1}(\theta)=0$ are handled by reparametrizing $\xi_j=g_j(\theta)$; the limit is $\chi^2(\dim\Omega-\dim\Omega_0)$.

Example 7.13 (Independence in a contingency table)

Counts X_{ij} of an $r\times c$ table are Multi$(n,(p_{ij}))$; $H_0: p_{ij}=p_{i\cdot}p_{\cdot j}$. MLEs: $\hat p_{ij}=x_{ij}/n$, and $\hat p_{ij}^0=\frac{x_{i\cdot}}{n}\frac{x_{\cdot j}}{n}$ under H_0:

$$2\{l(\hat p)-l(\hat p^0)\}=2\sum_{i,j}X_{ij}\log\frac{X_{ij}}{n\hat p_{i\cdot}\hat p_{\cdot j}}.$$

$\dim\Omega=rc-1$, $\dim\Omega_0=(r-1)+(c-1)$, so the limit is $\chi^2((r-1)(c-1))$. Pearson's statistic with estimated marginals gives the **chi-square test of independence**:

$$\sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(O_{ij}-\hat E_{ij})^2}{\hat E_{ij}}\ge\chi^2_\alpha((r-1)(c-1)),\qquad O_{ij}=X_{ij},\ \hat E_{ij}=\frac{X_{i\cdot}X_{\cdot j}}{n}.$$

Example 7.14 (A worked 2×3 table)

200 students by year and preferred study place:

	library	home	cafe	total
lower	40	30	30	100
upper	20	50	30	100
total	60	80	60	200

$\hat E_{1j}=\hat E_{2j}=(30,40,30)$ and

$$2\Big(\frac{10^2}{30}+\frac{10^2}{40}+0\Big)=11.67>\chi^2_{0.05}(2)=5.99,$$

reject independence at 5%.

Non-identically distributed observations. Under suitable conditions the results hold with $\bar I_n(\theta)=\frac1nE_\theta[-\ddot l(\theta)]$ for $I(\theta)$: $W_n=n(\hat\theta-\theta_0)'\bar I_n(\theta_0)(\hat\theta-\theta_0)$, $R_n=n\,\dot l(\theta_0)'\bar I_n(\theta_0)^{-1}\dot l(\theta_0)$.

Example 7.15 (Homogeneity of multinomial populations)

Independent $\mathbf{X}_i\sim$ Multi$(n_i,\mathbf{p}_i)$, $i\le r$, c cells; $H_0:\mathbf{p}_1=\cdots=\mathbf{p}_r$, pooled $\hat p_j^0=X_{\cdot j}/n$. With $O_{ij}=X_{ij}$, $\hat E_{ij}=n_i\hat p_j^0$, the LRT statistic $2\sum O_{ij}\log(O_{ij}/\hat E_{ij})$ is approximately Pearson's $\sum\sum(O_{ij}-\hat E_{ij})^2/\hat E_{ij}$ ($2u\log u\approx2(u-1)+(u-1)^2$ near $u=1$), with $r(c-1)-(c-1)=(r-1)(c-1)$ d.f., as for independence though row totals are fixed. (The limit follows from a quadratic form in $\sqrt{n_i}(\hat{\mathbf{p}}_i-\hat{\mathbf{p}}^0)$ with an idempotent matrix of rank $(r-1)(c-1)$.)

7.4 Selected exercises

Motivation. Book exercises: new LRTs, power of the t test, and the Poisson homogeneity statistic as Pearson's χ^2.

Exercise 7.6 (Power of the two-sided t test)

Let $X_1, \dots, X_n$ ($n \geq 2$) be $N(\mu, \sigma^2)$ and consider the two-sided t test of Example 7.5 with statistic $t = (\bar{X} - \mu_0)/(S/\sqrt{n})$.
(a) Show that $t \overset{d}{=} (Z + \delta)/\sqrt{V/(n-1)}$, where $\delta = (\mu - \mu_0)/(\sigma/\sqrt{n})$, $Z \sim N(0,1)$, $V \sim \chi^2(n-1)$, and Z, V are independent.
(b) Show that the power $P_{\mu,\sigma^2}\{|t| \geq t_{\alpha/2}(n-1)\}$ is a function of $|\delta|$ alone and is increasing in $|\delta|$.

Solution.

(a) With $Z = (\bar{X}-\mu)/(\sigma/\sqrt{n})$ and $V = (n-1)S^2/\sigma^2$, independent $N(0,1)$ and $\chi^2(n-1)$ (Chapter 4), dividing by $\sigma/\sqrt{n}$:

$$t = \frac{(\bar{X} - \mu)/(\sigma/\sqrt{n}) + (\mu - \mu_0)/(\sigma/\sqrt{n})}{S/\sigma} = \frac{Z + \delta}{\sqrt{V/(n-1)}}.$$

(b)

Step 1. With $c = t_{\alpha/2}(n-1)$ and $g_k(\delta) = P(|Z + \delta| \geq k) = 1 - \Phi(k - \delta) + \Phi(-k - \delta)$, by independence

$$\gamma(\delta) = P(|t| \geq c) = E\big[g_K(\delta)\big], \qquad K = c\sqrt{V/(n-1)}.$$

Step 2. $-Z \overset{d}{=} Z$ gives $g_k(-\delta) = g_k(\delta)$, so γ depends on $|\delta|$.

Step 3. $g_k'(\delta) = \phi(k - \delta) - \phi(k + \delta) > 0$ for $\delta > 0$ (as $|k - \delta| < k + \delta$).

Step 4. $K > 0$ a.s., so $g_K(\delta_1) < g_K(\delta_2)$ for $0 \leq \delta_1 < \delta_2$; take expectations. ($t$ is *noncentral* t with noncentrality δ.) ■

Exercise 7.9 (A one-sided test for Beta$(\theta, 1)$)

Let $X_1, \dots, X_n$ have density $f(x;\theta) = \theta x^{\theta-1}$ on $(0,1)$, $\theta > 0$. Find the level-α LRT of $H_0 : \theta \leq 1$ against $H_1 : \theta > 1$ ($0 < \alpha < \frac{1}{2}$).

Solution.

Step 1. With $T = -\sum_i \log X_i > 0$, $l(\theta) = n \log \theta - (\theta - 1)T$ is strictly concave, maximal at $\hat{\theta} = n/T$; over $(0, 1]$ the MLE is $\min(\hat{\theta}, 1)$.

Step 2. For $\hat{\theta} \leq 1$ the statistic is 0; for $\hat{\theta} > 1$ ($T < n$)

$$2\{l(\hat{\theta}) - l(1)\} = 2\{n \log(n/T) - n + T\} =: h(T), \qquad h'(T) = 2\Big(1 - \frac{n}{T}\Big) < 0.$$

h decreases from $+\infty$ to 0 on $(0, n)$, so the LRT rejects for $T \leq d$, $d \in (0, n)$.

Step 3. $P(-\log X_i > s) = e^{-\theta s}$, so $2\theta T \sim \chi^2(2n)$ and $P_\theta(T \leq d) = P\{\chi^2(2n) \leq 2\theta d\}$ increases in θ: size at $\theta = 1$.

Step 4. So reject iff

$$-2\sum_{i=1}^{n} \log X_i \leq \chi^2_{1-\alpha}(2n),$$

the lower α point. As $\alpha < \frac{1}{2}$ it lies below the median of $\chi^2(2n)$, which is below the mean $2n$ (gamma median $<$ mean), so $d < n$ as needed. ■

Exercise 7.15(a) (Testing the endpoint of a uniform law)

Let $X_1, \dots, X_n$ be $U[0, \theta]$, $\theta > 0$. Find the level-α LRT of $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$.

Solution.

Step 1. With $M = \max_i X_i$, $L(\theta) = \theta^{-n} I(\theta \geq M)$, $\hat{\theta} = M$, and

$$\frac{L(\theta_0)}{L(\hat{\theta})} = \begin{cases} (M/\theta_0)^n, & M \leq \theta_0, \\ 0, & M > \theta_0. \end{cases}$$

Reject when the ratio is small: $M > \theta_0$ or $M \leq k\theta_0$, $k \in (0, 1)$.

Step 2. Under H_0, $P(M > \theta_0) = 0$, $P(M \leq k\theta_0) = k^n = \alpha$:

$$\text{reject } H_0 \iff M > \theta_0 \text{ or } M \leq \alpha^{1/n}\theta_0,$$

of size exactly α. Unlike Section 7.3, $2\{l(\hat{\theta}) - l(\theta_0)\} = -2n \log(M/\theta_0)$ is exactly $\chi^2(2)$ ($P(M/\theta_0 \leq e^{-s/n}) = e^{-s}$). ■

Exercise 7.17(a)(b) (Homogeneity of Poisson means)

Let $X_{i1}, \dots, X_{in_i}$ be a sample from Poisson(θ_i), $i = 1, \dots, k$ ($k \geq 2$), the samples being independent. Put $X_{i\cdot} = \sum_j X_{ij}$, $n = n_1 + \cdots + n_k$, $\hat{\theta}_i = X_{i\cdot}/n_i$ and $\hat{\theta}^0 = \sum_i X_{i\cdot}/n$. For $H_0 : \theta_1 = \cdots = \theta_k$:
(a) Show that the LRT statistic is $2\sum_i n_i\{\hat{\theta}_i \log(\hat{\theta}_i/\hat{\theta}^0) - \hat{\theta}_i + \hat{\theta}^0\}$ (with $0 \log 0 = 0$).
(b) Show that if $n_i \to \infty$ with $n_i/n \to \gamma_i \in (0, 1)$, then under H_0 (with common value $\theta > 0$) the statistic is approximately $\sum_i (X_{i\cdot} - n_i\hat{\theta}^0)^2/(n_i\hat{\theta}^0)$, in the sense that the ratio of the two tends to 1 in probability.

Solution.

(a) Up to a constant, $l(\theta) = \sum_i \{X_{i\cdot} \log \theta_i - n_i\theta_i\}$, maximized at $\hat{\theta}_i$, and under H_0 at $\hat{\theta}^0$. With $X_{i\cdot} = n_i\hat{\theta}_i$,

$$2\{l(\hat{\theta}) - l(\hat{\theta}^0)\} = 2\sum_i \{n_i\hat{\theta}_i \log(\hat{\theta}_i/\hat{\theta}^0) - n_i\hat{\theta}_i + n_i\hat{\theta}^0\}.$$

(The linear terms sum to 0.)

(b)

Step 1. For $a > 0$, $g_a(x) = x \log(x/a) - x + a$ has $g_a(a) = g_a'(a) = 0$, $g_a''(x) = 1/x$; by Taylor, for $x > 0$

$$g_a(x) = \frac{(x-a)^2}{2\xi}, \qquad \xi \text{ between } x \text{ and } a.$$

Step 2. Under H_0, $\hat{\theta}_i, \hat{\theta}^0 \xrightarrow{p} \theta$ (LLN), so $P(\hat{\theta}_i > 0) \to 1$, $\xi_i \xrightarrow{p} \theta$, and

$$2\{l(\hat{\theta}) - l(\hat{\theta}^0)\} = \sum_i n_i \frac{(\hat{\theta}_i - \hat{\theta}^0)^2}{\hat{\theta}^0} \cdot \frac{\hat{\theta}^0}{\xi_i},$$

with $\hat{\theta}^0/\xi_i \xrightarrow{p} 1$.

Step 3. $\sum_i a_i b_i / \sum_i a_i$ lies between $\min_i b_i$ and $\max_i b_i$ ($a_i \geq 0$), so it $\xrightarrow{p} 1$ for fixed k. With $n_i(\hat{\theta}_i - \hat{\theta}^0)^2/\hat{\theta}^0 = (X_{i\cdot} - n_i\hat{\theta}^0)^2/(n_i\hat{\theta}^0)$ this is the claim (limit $\chi^2(k-1)$, Theorem 7.2). ■